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|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------|
| Areas of Interest         | <ul style="list-style-type: none"> <li>Econometrics, Development Economics, Health Economics, Environmental Economics, Machine Learning, Game Theory.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                        |             |
| Qualification Highlights  | <ul style="list-style-type: none"> <li>Graduate in <b>Statistics with Development Economics, Econometrics, Real Analysis, Linear Models</b>, and <b>Advanced Time Series</b> courses. 5+ years of research experience specializing in <b>RCT design</b>, <b>data analysis</b>, <b>econometrics</b> and behavioral economics</li> <li>Proficient in <b>R, Python, Stata, SAS, C++</b>, <b>SQL, HTML, CSS</b>, <b>oTree</b>, and <b>MATLAB</b>.</li> <li>Experienced in <b>Data Mining, Visualization, Large Data Analysis Execution, Results Interpretation</b>, and <b>Report Writing</b>. Strong skills in <b>critical thinking, communication</b>, and <b>collaboration</b> with diverse backgrounds.</li> <li>Excellent <b>organizational skills</b> with the ability to <b>manage multiple projects</b>. Demonstrated <b>attention to detail</b> in all aspects of work. Capable of <b>working independently</b> and utilizing various <b>scripting languages</b> for data manipulation.</li> </ul> |                        |             |
| Education and Test Scores | <b>Degree</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Institution</b>     | <b>Year</b> |
|                           | B.Sc.(Hons) in Statistics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | University of Calcutta | 2015        |
|                           | M.Sc. in Statistics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | University of Calcutta | 2017        |
|                           | PhD in Economics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | UC Riverside           | 2025-2031   |
|                           | <b>GRE Score:</b> Quantitative: 170, Verbal: 158                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                        |             |
|                           | <b>IIT JAM Rank:</b> All India Rank 86                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |             |
| Certificate Courses       | <b>Course Name and Institution</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                        | <b>Year</b> |
|                           | Managing Successful Field Research (MSFR), World Bank                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                        | 2024        |
|                           | Macroeconometric Forecasting/Macroeconomics for Climate Change, IMF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        | 2024        |

## Research and Publications

- R-package (DHSr):** R-package to facilitate developing regression summary statistics datasets from repeated nesetd regressions on large datasets (e.g., DHS) and a novel panel/spatial-panel econometric clustering tool [\[CRAN Publication\]](#) [\[CRAN Documentation\]](#) [\[ Github Documentation/Workflow\]](#)
- Community Effect on Educational Participation in Contemporary India: A District Level Analysis(Working Paper):** I address the limitations in my previous work by developing district-level mixed-effects regression models to analyze educational participation in India. The study examines variations in effect of gender, caste, and religion across Indian districts, accounting for inter-district spillover, and uses Markov Chain Analysis to assess community effect on education, and role of wealth. Most importantly, it stresses the need to take regional variations into account. [\[ Link: Draft Paper\]](#)

## Other Research:

- Multidimensional Poverty Analysis: assessing poverty patterns among different demographic groups of India:** (Published at Journal of Progressive Research in Social Sciences Vol 13) [\[JPRSS\]](#)
- Two Case Studies: Statistically Observable Data for LGBTQI+ Individuals in the Indian Subcontinent**
  - Non-parametric analysis on transgender individuals matched with their most similar male/female counterparts from the same households to assess the effect of being transgender on education and health deprivation
  - Comparing scores for GAD-7, PHQ-9, and GHQ-12 mental health indicators between LGBTQI+ students and a control group of heterosexual students of the same age from 28 different Indian cities, during the COVID-19 pandemic
- MSc Dissertations:**
  - Time Series and Regression Analysis of Measles Cases (1928-1972):** Applied ARIMA and GARCH models for prediction and used VAR and SEM models to capture instantaneous causality.
  - Randomized Response Technique for Multiple Categories:** Developed a generalized RRT for populations with multiple categories to reduce bias in survey responses.

## Academic Work Experience and Collaborations

| Institution                                              | Position                       | Responsibilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cambridge/Kingston University<br>PI: Prof. Jalal Siddiki | Research Assistant & Co-author | <ol style="list-style-type: none"> <li>We measure <b>home ownership segregation</b> within specific <b>travel-to-work areas</b>, where the level of diversity in home ownership is represented by an <b>entropy index</b>.</li> <li>Research objective is to assess the influence of home ownership segregation on <b>inter-generational upward mobility</b> using <b>historical and cross-sectional data</b> from the <b>European Central Bank</b>.</li> <li>I am supervising the <b>empirical section - writing and coding (Stata)</b>.</li> </ol> |

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|--------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indian Institute of Management, Ahmadabad<br>PI: Vinay Kumar Jha (PhD student) | Programmer/Associate         | <b>Project 1:</b> Designing a <b>lab-in-field</b> experiment to evaluate a <b>novel matching mechanism</b> for assigning Indian bureaucrats, comparing it to the current method, and testing empirical versus theoretical findings on equilibrium and strategy-proof properties. Tools used: R, Python, HTML, oTree; Survey Population: UG students/job aspirants. <b>Project 2:</b> Assisted to create an ideal dataset for <b>NGO Swadhaar FinAccess</b> for social-network/peer-effect analysis among individuals near the poverty line in India. |
| University of Calcutta<br>Supervisor: Prof. Sugata Sen Roy                     | Graduate Research Apprentice | <b>1.</b> Computed <b>Multidimensional Poverty Index</b> using <b>IPUMS DHS data</b> . <b>2.</b> Applied analytical tools to assess government policies, introducing neural networks and unsupervised methods. <b>3.</b> Adapted a <b>Cluster Analysis-based Missing Data imputation method</b> . <b>4.</b> Conducted literature reviews and academic writing for research publications. <b>5.</b> Performed literature reviews of public finance literature to assist senior fellows' research.                                                     |

## Industry Work Experience

| Organization                  | Designation and Duration                                  | Job Deliverables                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GSK/Cytel                     | Senior Data Analyst<br>(May 2024 - Present)               | <b>1.</b> Lead <b>Health Economist/Data Analyst</b> for a <b>Clinical Trial</b> to research efficacy of a new <b>respiratory disease medication</b> , optimizing <b>Carbon Footprint</b> reduction. <b>2.</b> Developing <b>R packages/Dashboard</b> for data analysis.                                                                                                                                                                                                                                                                           |
| IQVIA Biotech                 | Statistical Programmer II<br>(July 2022 – April 2024)     | <b>1.</b> Developed and documented programs for <b>Phase 3 clinical trials</b> , adhering to <b>FDA guidelines</b> . <b>2.</b> Interacted with global <b>stakeholders</b> across the <b>EU, USA, and Japan</b> . <b>3.</b> Enhanced process <b>quality and efficiency</b> using <b>AI and ML</b> . <b>4.</b> Trained <b>group members</b> and <b>field-enumerators</b> on new processes and <b>Electronic Data Capture</b> tools. <b>5.</b> Managed multiple <b>projects simultaneously</b> . Awards: <b>Excellence in Data Innovation 2023</b> . |
| Clinipace Clinical Research   | Senior Statistical Programmer<br>(Sep 2021 – June 2022)   | <b>1.</b> Lead <b>Statistician</b> for multiple <b>randomized controlled trials</b> . <b>2.</b> Designed and modified <b>Randomized Controlled Trial</b> . <b>3.</b> Created complex <b>statistical tables and graphs</b> . Utilized <b>MATLAB</b> for various analyses.                                                                                                                                                                                                                                                                          |
| Novartis Healthcare Pvt. Ltd. | Statistical Programmer<br>(Apr 2019 – June 2021)          | <b>1.</b> Lead <b>SAS Statistical Programmer</b> for <b>clinical trials</b> . <b>2.</b> Designed and modified <b>clinical trials</b> . <b>3.</b> Collaborated globally on <b>data quality</b> . <b>4.</b> SME for <b>AI-based clinical programming</b> . <b>5.</b> Utilized scripting for <b>automation</b> . Awards: <b>Best SME 2020, IMPACT Performer 2019</b> .                                                                                                                                                                               |
| Novartis Healthcare Pvt. Ltd. | Associate Statistical Programmer<br>(Aug 2017 – Apr 2019) | <b>1.</b> Cleaned and analyzed <b>clinical trial datasets</b> using <b>SAS and R</b> . <b>2.</b> Ensured robustness in <b>P21 checks</b> and developed outputs. <b>3.</b> Organized <b>departmental activities</b> . <b>4.</b> Implemented knowledge in <b>clinical data standards</b> . <b>5.</b> Edited and finalized <b>clinical reports</b> .                                                                                                                                                                                                 |

- **Recent Academic Award:** Recipient of the Dean's Distinguished Fellowship (2025 PhD cohort) at University of California, Riverside.
- **Conference Volunteering:** Participated in the "Ninth International Triennial Calcutta Symposium on Probability and Statistics", organized by the Department of Statistics, University of Calcutta.
- **Pre-College Academic Achievement:** Recipient of a State-level Scholarship for Excellence in Secondary Exam (Awarded by the Government of India to the top 1% of students).
- **PhD Thesis Assistance:**
  - Assisted Dr. Kunal Datta in his PhD thesis "Mixed-halide perovskite semiconductors for multijunction photovoltaics" – submitted to TU Eindhoven in The Netherlands.
  - Supported Dr. Saurav Ghosh (University of Calcutta) in his paper "Prevalence of Work-related Musculoskeletal Disorders in Traffic Police Personnel in Maintaining Transportation: A Cross-sectional Study" (submitted to the Journal of Transport and Health).

## Additional Courses

- Advanced Linear Algebra, UT Austin
- Stanford University CS229 (Machine Learning), CS230 (Deep Learning), CS231 (Deep Learning for Computer Vision)
- Microeconomic Theory I-IV (graduation level Microeconomics course 14.121, 14.122, 14.123, 14.124) - MIT
- Spatial Data Analysis course by Prof. Michael J. Pyrcz (University of Texas)

## Personal Information

**Language Proficiency:** English, Bengali, Hindi

**Base SAS:** 9.4 **Other Software Knowledge:** R, Python, MATLAB, ArcGIS, QGIS, STATA, MINITAB

I hereby declare that the above information is correct to the best of my knowledge.

Date: 19-Oct-2024

Place: Kolkata

**Arnab Samanta**